## Homework

**DM11 - A face and its altitude - 50 min.**

In an orthonormal coordinate system with origin $O$, consider the points $A(2;0;0)$, $B(0;2;0)$ and $C(0;0;2)$.

1. Prove that the triangle $ABC$ is equilateral and calculate its area.

2. Show that the vector $\vec{n} = (1;1;1)$ is normal to the plane $(ABC)$.

3. Let $H$ the orthogonal projection of $O$ on $(ABC)$. Justify that its coordinates are of the form $(t;t;t)$. Determine $t$ using $\overrightarrow{AH}\cdot\vec{n}=0$.

4. Calculate $OH$, then the volume of the tetrahedron $OABC$ by taking $ABC$ as base. Check this volume by taking $OAB$ as base.